

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD
CAMPUS
EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2025-26

TUTORIAL- 5

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Controlled Rectifiers (Phase Controlled, Converter)

- 1) A single phase fully controlled half wave converter is used for controlling the illumination of an electric bulb of rating 230V 100W. If the source provided is 230V, 50Hz, determine the parameters for $\alpha = 0, 30, 60, 90, 120, 150, 180$ degrees.
- 2) A single phase fully controlled full bridge rectifier is supplied at 120V. If there is a voltage drop of 1.5 V across each thyristor, then determine the average load voltage for firing delay angles of $0^\circ, 45^\circ$ and 90° . Assume continuous and ripple free load current. Also determine the firing angle at which the average load voltage will be equal to zero. Find the maximum peak inverse voltage of the thyristor.
- 3) A 200V, 10A, 1500rpm, separately excited DC machine has an armature resistance 0.5Ω and motor inductance of 50mH. The motor is supplied from a single phase fully controlled bridge converter operating from a 230V, 50Hz single phase supply with a firing angle of 30 degrees. At what speed motor will supply full load torque at 30-degree firing (assume continuous current conduction mode of operation).
- 4) A 230 V, 50 Hz, one-pulse Thyristor controlled rectifier is triggered at a firing angle of 40° and load current extinguishes at an angle of 210° . Find the turn off time, average output voltage and the average load current for
 - (a) $R = 5\Omega$ and $L = 2\text{ mH}$.
 - (b) $R = 5\Omega$, $L = 2\text{ mH}$ and $E = 110\text{ V}$.
- 5) A single-phase full converter bridge is connected to RLE load. The source voltage is 230 V, 50 Hz. The average load current of 10 A is constant over the working range. For $R = 0.4\Omega$ and $L = 2\text{mH}$, compute:
 - (a) Firing angle delay for $E = 120\text{ V}$,
 - (b) Firing angle delay for $E = -120\text{ V}$.Indicate which source is delivering power to load in parts (a) and (b).
- (c) In case output current is assumed constant, find the input P.F for both parts (a) and (b).